

MLOps — Advanced Full-Length Exam with Answers

PART A — Multiple Choice Questions (with Explanations)

Q1. The key bottleneck MLOps aims to remove compared to traditional ML workflows is:

- A. Low model accuracy
- B. Manual deployment and integration gaps
- C. Lack of hyperparameter tuning
- D. Limited visualization tools

✓ **Answer:** B → MLOps automates the continuous testing, packaging, and deployment pipelines.

Other options are model-level, not system-level problems.

Q2. In MLflow, which of the following is stored under the “artifacts” folder?

- A. Metrics log
- B. Hyperparameters
- C. Trained model files or visual plots
- D. Experiment name

✓ **Answer:** C → Artifacts store output files like models, charts, or confusion matrices.

A & B are logged separately, *D* is metadata.

Q3. Which Kubernetes component ensures that a certain number of pod replicas are always running?

- A. Node Controller
- B. ReplicaSet
- C. ConfigMap
- D. Service

✓ **Answer:** B → ReplicaSet maintains the desired number of pods.

A manages nodes; *C* holds configs; *D* exposes network endpoints.

Q4. During concept drift, which metric will typically drop first?

- A. Precision
- B. Recall
- C. F1-score
- D. Accuracy

✓ **Answer:** D → Accuracy drops broadly as predicted vs actual labels misalign due to changing relationships.

Precision/Recall may vary by class but not necessarily together.

Q5. In a Conda YAML file, pip dependencies are included under which key?

- A. `requirements:`
- B. `pip:`
- C. `libraries:`
- D. `dependencies:`

✓ **Answer:** B → The pip dependencies appear nested under - `pip:`.

A & C are invalid YAML keys; D lists Conda packages.

Q6. In Docker, every command like RUN or COPY creates a new:

- A. Base image
- B. Container
- C. Layer
- D. Registry

✓ **Answer:** C → Each instruction adds a new filesystem layer; this makes caching efficient.

Base image is only defined by `FROM`.

Q7. The default MLflow backend storage is:

- A. PostgreSQL
- B. FileStore (local directory)
- C. SQLite cloud DB
- D. Redis cache

✓ **Answer:** B → By default, MLflow logs to `./mlruns` directory (FileStore).

PostgreSQL is for advanced remote tracking.

Q8. Horizontal scaling in Kubernetes helps when:

- A. A model needs larger RAM
- B. Multiple concurrent API requests arrive
- C. A single GPU job grows in size
- D. Disk I/O bottlenecks occur

✓ **Answer:** B → Horizontal scaling replicates pods to handle more simultaneous traffic.
A = vertical scaling; C requires bigger node; D is storage-level issue.

Q9. Which AutoML system supports declarative configuration via YAML and handles tabular, text, and image data?

- A. Ludwig
- B. FLAML
- C. H2O AutoML
- D. AutoKeras

✓ **Answer:** A → Ludwig uses YAML-based configs and supports multiple modalities.
FLAML is code-based and lighter; H2O and AutoKeras differ architecturally.

Q10. In CI/CD, unit tests for ML should verify:

- A. GPU utilization
- B. Training dataset schema
- C. Model's ROC curve
- D. UI responsiveness

✓ **Answer:** B → Validating schema prevents silent data corruption across updates.
A, C, D are performance or UI concerns.

Q11. Which of the following best prevents “train-serve skew”?

- A. Early stopping
- B. Shared feature store
- C. Using random seeds
- D. Model checkpoints

✓ **Answer:** B → Ensures both training and inference use identical feature transformations.

Q12. Docker's ENTRYPOINT is primarily used to:

- A. Run commands during build
- B. Define container startup behavior
- C. Set environment variables
- D. Copy files from host

✓ **Answer:** B → ENTRYPOINT defines the executable command at runtime.
RUN executes during image build, *CMD* provides default args.

Q13. The Kubernetes Service object provides:

- A. Container logs
- B. Network abstraction for pods
- C. Model monitoring
- D. Job orchestration

✓ **Answer:** B → A Service exposes pod endpoints internally or externally.
D = Jobs/Deployments handle orchestration.

Q14. Which MLflow function logs multiple metrics at once?

- A. `mlflow.log_metrics()`
- B. `mlflow.log_all()`
- C. `mlflow.batch_metrics()`
- D. `mlflow.log_batch()`

✓ **Answer:** A → Accepts a dictionary of key-value metric pairs.

Q15. "Drift detection" should be done at which stage of the MLOps lifecycle?

- A. Data ingestion
- B. Post-deployment monitoring
- C. Model training
- D. During version control

✓ **Answer:** B → Drift is detected in live environments once models are serving predictions.

Q16. The command `docker push myrepo/mlapi:v2` does what?

- A. Pushes image to Docker Hub
- B. Saves local copy

- C. Starts a new container
- D. Commits code

✓ **Answer:** A → Uploads tagged image to configured registry repository.

Q17. In SageMaker, “Model Monitor” is used for:

- A. Debugging EC2 latency
- B. Checking drift and data quality
- C. Fine-tuning hyperparameters
- D. Logging user sessions

✓ **Answer:** B → Detects anomalies or drift between training and production data.

Q18. The core function of a model registry in MLOps is to:

- A. Train models
- B. Manage labeled data
- C. Store and version approved models
- D. Run CI pipelines

✓ **Answer:** C → It acts like a GitHub for model binaries and metadata.

Q19. The Dockerfile line `RUN pip install -r requirements.txt` executes:

- A. At runtime
- B. During build, creating a cached layer
- C. In deployed container
- D. Only if user triggers it manually

✓ **Answer:** B → RUN executes at build time and creates a frozen dependency layer.

Q20. FLAML stands for:

- A. Fast Light AutoML
- B. Flexible Learning AutoML
- C. Fully Labeled Automated ML
- D. Federated Learning for ML

✓ **Answer:** A → Designed by Microsoft to provide cost-efficient AutoML optimization.

Q21. Which CI/CD step would fail first if your Dockerfile has a missing dependency?

- A. Build stage
- B. Deploy stage
- C. Test stage
- D. Monitor stage

✓ **Answer:** A → Missing dependency causes image build failure.

Q22. In Feature Stores, “online” vs “offline” features differ by:

- A. Data freshness
- B. File format
- C. Storage engine
- D. Access control

✓ **Answer:** A → Online = real-time serving; Offline = batch historical data.

Q23. Which cloud ML service allows TPU-based training jobs?

- A. AWS SageMaker
- B. Azure ML
- C. GCP Vertex AI
- D. IBM Watson Studio

✓ **Answer:** C → Vertex AI integrates directly with Google TPUs.

Q24. The Conda flag `--no-builds` ensures:

- A. Faster installation
- B. Cross-platform compatibility by removing OS-specific info
- C. GPU support
- D. Removal of pip packages

✓ **Answer:** B → Exports environment independent of system architecture.

Q25. CI/CD is most valuable for which MLOps challenge?

- A. Hyperparameter optimization
- B. Continuous retraining and version deployment
- C. GPU scheduling
- D. Data visualization

✓ **Answer:** B → Enables automatic retraining and rollout when new data arrives.

Q26. A Docker image is composed of:

- A. Single executable file
- B. Union of read-only layers
- C. Kernel modules
- D. Virtual machine snapshot

✓ **Answer:** B → Each instruction adds a layer; together form the image filesystem.

Q27. MLflow's `set_tracking_uri()` can point to which types of backends?

- A. Only local folders
- B. SQL databases or HTTP servers
- C. Git repositories
- D. Docker registries

✓ **Answer:** B → Supports local FileStore or remote servers (e.g., PostgreSQL via REST API).

Q28. Concept drift detection often uses which signal?

- A. Feature variance
- B. Label distribution comparison over time
- C. API latency
- D. Number of users

✓ **Answer:** B → Labels changing pattern relative to inputs indicates conceptual shift.

Q29. Kubernetes' "Deployment" object mainly manages:

- A. Scaling and rolling updates of pods
- B. Data replication across nodes
- C. Logs aggregation
- D. Namespace authentication

✓ **Answer:** A → Handles rollout strategies and pod version control.

Q30. The command to rebuild environment from YAML:

- A. `conda recreate -f environment.yml`

- B. `conda env create -f environment.yml`
- C. `conda build env.yml`
- D. `conda import environment.yml`

✓ **Answer:** B → Creates identical environment from file.

Q31. In a FastAPI app, `uvicorn main:app --reload` does what?

- A. Starts the app with hot reload for dev mode
- B. Builds Docker image
- C. Updates Conda packages
- D. Logs MLflow metrics

✓ **Answer:** A → Uvicorn runs ASGI app with auto restart on code change.

Q32. In Vertex AI, “Feature Store” enables:

- A. Label engineering
- B. Reusable, versioned feature storage for training and serving
- C. Visualization of models
- D. TPU configuration

✓ **Answer:** B → Central repository for managed feature definitions.

Q33. The hypervisor in a VM acts as:

- A. Hardware abstraction layer
- B. Shared kernel provider
- C. Virtual disk image
- D. Networking daemon

✓ **Answer:** A → It virtualizes CPU and memory for multiple guest OSs.

Q34. Which AWS service complements SageMaker for data preprocessing?

- A. AWS Glue
- B. AWS Rekognition
- C. AWS CloudWatch
- D. AWS Kinesis

✓ **Answer:** A → Glue performs ETL before SageMaker training.

Q35. MLflow is framework-agnostic because:

- A. It re-implements all frameworks internally
- B. It uses Python's pickle
- C. It provides generic APIs for any language or library
- D. It runs only on TensorFlow

✓ **Answer:** C → Supports sklearn, XGBoost, PyTorch, etc., through unified API.

Q36. The command `docker exec -it container_id bash` is used to:

- A. Start container
- B. Enter shell inside a running container
- C. Stop container
- D. Build container image

✓ **Answer:** B → Opens interactive bash session within container.

Q37. What triggers retraining in automated MLOps pipelines?

- A. Data drift alert or drop in accuracy metrics
- B. Code formatting changes
- C. Container rebuild
- D. RAM increase

✓ **Answer:** A → Monitors detect drift and schedule retraining via CI/CD.

Q38. In Conda, `conda list` command displays:

- A. Package history
- B. All installed packages in current env
- C. Available updates
- D. YAML file content

✓ **Answer:** B → Lists installed packages with version numbers.

Q39. The FROM instruction in Docker must always appear:

- A. At the end of Dockerfile
- B. At the start (first instruction)

- C. Before COPY
- D. Only if ENTRYPOINT is defined

✓ **Answer:** B → It defines the base image foundation for all subsequent layers.

Q40. The advantage of microservice architecture for ML APIs is:

- A. Tighter coupling
- B. Independent scaling and easier fault isolation
- C. Single monolithic control
- D. Lower latency always

✓ **Answer:** B → Allows independent updates, scaling, and recovery.

Q41. A YAML environment file that includes `name :` at top indicates:

- A. Conda's base version
- B. The environment's name identifier
- C. Pip's dependency list
- D. A runtime variable

✓ **Answer:** B → Specifies environment label for creation.

Q42. In Docker, registry authentication errors occur when:

- A. Image tags mismatch
- B. Network policy blocks 443
- C. You try pushing without login credentials
- D. Container is stopped

✓ **Answer:** C → You must `docker login` before push.

Q43. The CI/CD build artifact for model deployment is usually:

- A. Model pickle
- B. Docker image
- C. YAML file
- D. Log file

✓ **Answer:** B → The Docker image packages model + dependencies.

Q44. MLflow's Projects component enables:

- A. Model serving
- B. Reproducible packaging of ML code
- C. Visualization
- D. CI testing

✓ **Answer:** B → Encapsulates code and environment for repeatable runs.

Q45. "Vertex AI Workbench" primarily provides:

- A. GPU provisioning
- B. Managed Jupyter notebooks integrated with pipelines
- C. AutoML dashboard
- D. Feature monitoring

✓ **Answer:** B → Combines BigQuery + Kubeflow notebooks.

Q46. Which of the following is NOT part of MLOps lifecycle?

- A. Data ingestion
- B. Model evaluation
- C. Marketing analysis
- D. Continuous monitoring

✓ **Answer:** C → It's business, not MLOps.

Q47. The command `docker stop $(docker ps -q)` will:

- A. Stop all running containers
- B. Delete images
- C. Remove dangling volumes
- D. Restart Docker

✓ **Answer:** A → `docker ps -q` lists IDs, `stop` halts them all.

Q48. Which ML monitoring metric detects sudden spike in input schema nulls?

- A. PSI
- B. Missing value ratio
- C. Accuracy
- D. Recall

✓ **Answer:** B → Detects feature quality drift.

Q49. CI/CD pipelines for ML commonly include which testing phase?

- A. Load testing for inference endpoints
- B. Browser rendering
- C. UI responsiveness
- D. Email alerts

✓ **Answer:** A → Ensures model APIs sustain traffic.

Q50. Conda environments differ from Docker containers because:

- A. Conda isolates system kernel too
- B. Conda only manages Python packages; Docker virtualizes the OS layer
- C. Conda is slower
- D. Docker doesn't support reproducibility

✓ **Answer:** B → Docker includes OS-level isolation; Conda does not.



PART B — Short Answer Questions (2–3 sentences)

1 Describe the difference between Dockerfile's RUN and CMD.

RUN executes during build and creates image layers. CMD runs at container startup and can be overridden by user command.

2 What happens if you ignore drift detection in a production ML model?

The model's predictions degrade over time because data and patterns evolve, causing business errors and loss of trust.

3 Explain why microservices are crucial for MLOps scalability.

Each model or service runs independently, enabling separate scaling, updates, and fault recovery without downtime.

4 Compare data drift and label drift.

Data drift = change in feature distribution; Label drift = shift in target output frequencies. Label drift especially impacts classification models.

5 Explain the role of the MLflow Tracking URI.

It specifies the backend (local folder or remote server) where all run metadata, parameters, and metrics are logged.

6 How does Kubernetes manage rolling updates for ML APIs?

It replaces pods incrementally, ensuring zero downtime by gradually routing traffic from old pods to new ones.

7 Why does CI/CD integrate with Docker in MLOps?

Because Docker images ensure identical environments across testing, staging, and production deployments.

8 What is the difference between Feature Store “online” and “offline” layers?

Offline layer supports batch training; Online layer serves real-time features during inference for low-latency APIs.

9 How can MLflow Model Registry prevent deployment errors?

By tracking versioned models and enforcing promotion workflows (Staging → Production) under governance.

10 Why is Conda environment YAML preferred over requirements.txt in ML projects?

It captures Python and non-Python dependencies plus channels, making complex environments reproducible across systems.

11 How is Vertex AI different from SageMaker?

Vertex AI natively integrates with TPUs and BigQuery; SageMaker is deeper integrated with AWS data and security stack.

12 Why is AutoML useful but not always optimal?

It automates model tuning but may not yield domain-optimized results; manual expert tuning can outperform it for niche data.

13 What does ‘hypervisor’ mean in virtualization context?

It’s a software layer managing multiple guest operating systems on the same hardware, allocating CPU and memory.

14 Describe one real-world cause of concept drift.

Changes in consumer behavior — e.g., post-pandemic shopping trends shift model assumptions about demand.

15 What are the benefits of using MLflow’s log_artifact()?

It stores custom outputs (plots, reports, models) within a structured experiment directory for reproducibility.

16 Why do containers start faster than VMs?

Containers reuse the host kernel, eliminating OS boot overhead, while VMs boot entire OS instances.